On the duality between contrastive and non-contrastive self-supervised learning

Quentin Garrido, Yubei Chen, Adrien Bardes, Laurent Najman, and Yann Lecun

ICLR 2023 best paper honorable mention

Presentor: Hojoon Lee

Contents

Preliminaries

- Taxonomy of Self-Supervised Learning.
- Taxonomy of Invariance Learning.

Duality

- Notations & Definitions.
- Why SimCLR is a Sample-Contrastive method?
- Why VICReg is a Dimension-Contrastive method?
- Equivalence of Sample-contrastive and Dimension-contrastive methods.
- Connections to SimSiam, BYOL, SwAV, DINO.

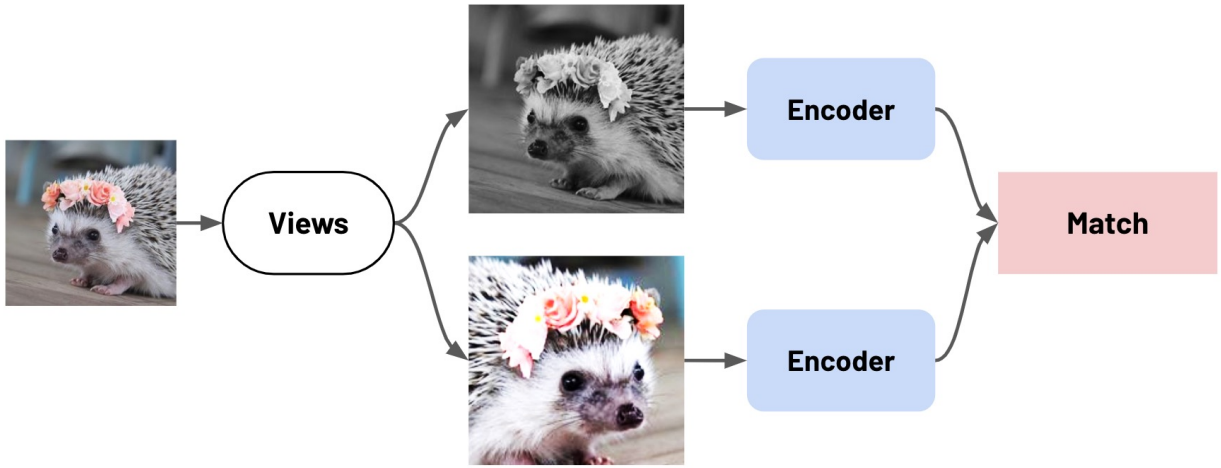
Future Work

- Integrating Invariance Learning and Generation.

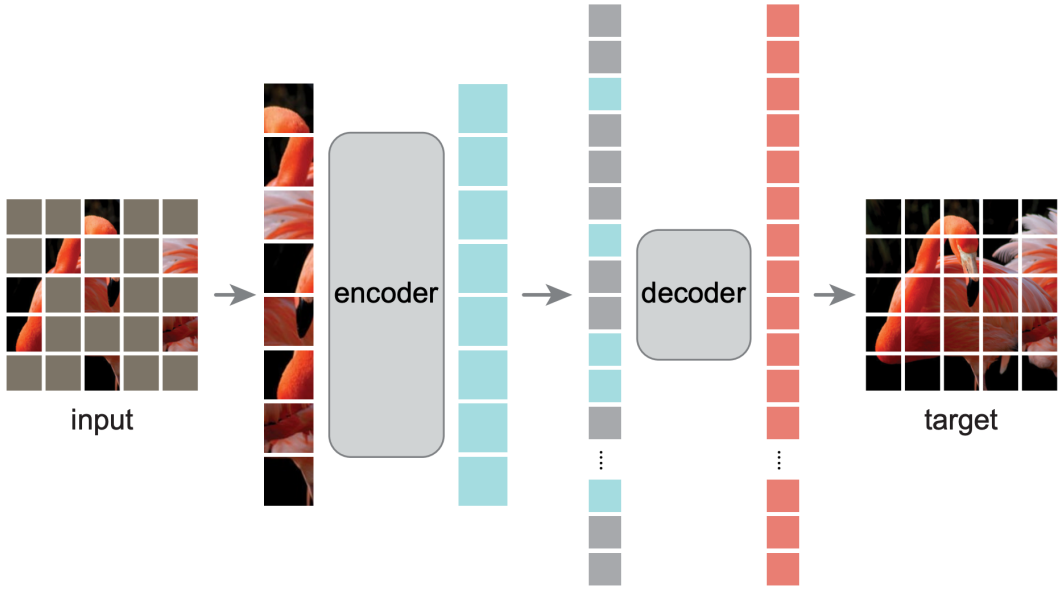
Taxonomy of Self-Supervised Learning

Invariance vs Generation

- Let X be data, $Z(X)$ be representation.
- Let I denotes mutual information of two random variables.
  - **Invariance:** Maximize $I(Z(X_1);Z(X_2))$ where X_1, X_2 are invariant data.
  - **Generation:** Maximize $I(Z(\bar{X});X)$ where $\bar{X}$ is perturbed version of X.



Invariance



Generative

Taxonomy of Invariance Learning

Core idea of Invariance Learning

- Representations of positive pairs should be similar.

Positive pair $f($ $) \approx f($ $)$

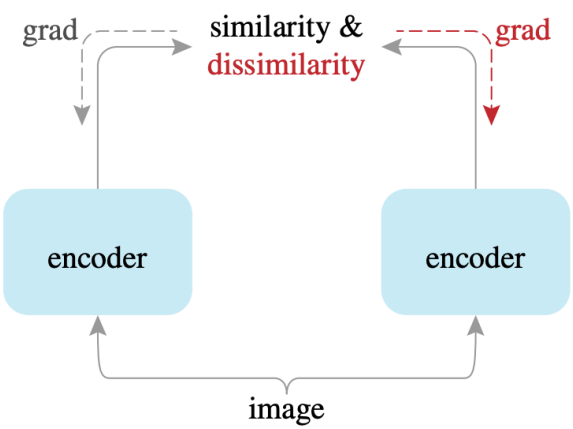
Problem of Invariance Learning: Mode Collapse

- Simply minimizing $|f($ $)-f($ $)\ |$ leads to mode collapse.
- Mode collapse: $\forall x,\ f(x)=c$

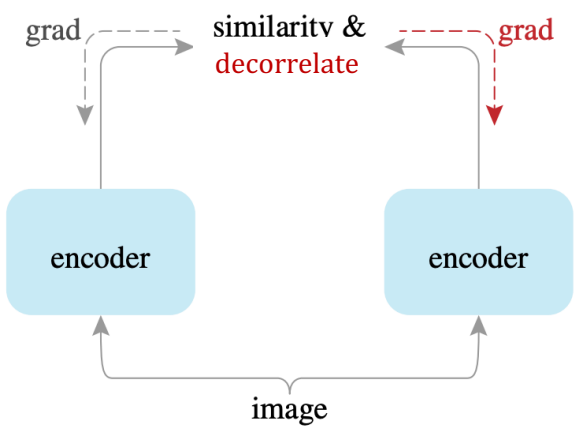
Taxonomy of Invariance Learning

Solutions for Mode Collapse in Invariance Learning

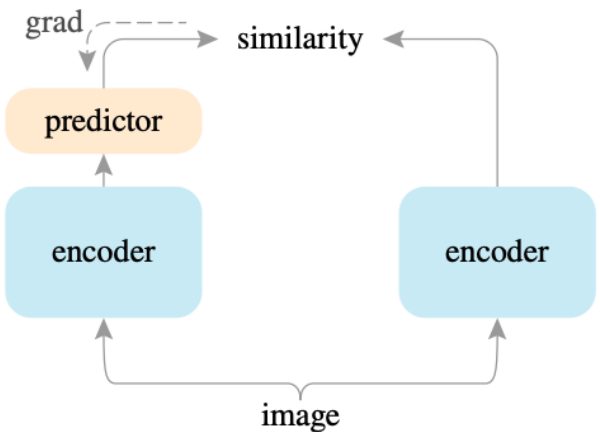
- Sample-Contrastive (SC): SimCLR, MoCo
  - Penalizing similarity between different pairs of images.
- Dimension-Contrastive (DC): VICReg, BarlowTwins
  - Penalizing similarity between different pairs of features.
- Non-Contrastive: SimSiam, BYOL, SwAV, DINO
  - Use asymmetric architecture.



Sample-Contrastive



Dimension-Contrastive



Non-Contrsative

Duality between sample-contrastive and dimension-contrastive self-supervised learning

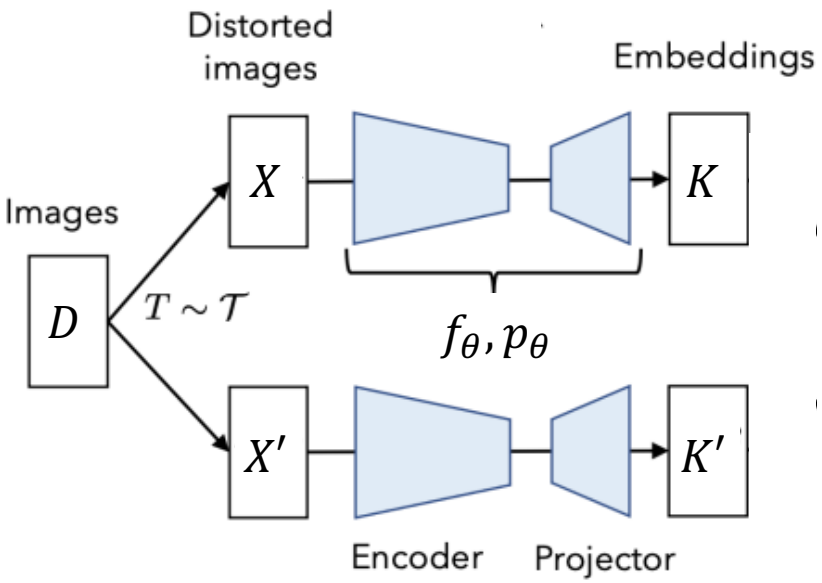
Notations & Definitions

Notations

- Batch size: N
- Embedding size: M

- Encoder (e.g., ResNet50): f_θ
- Projector (e.g., MLP): p_θ

- Data: $d_i \in R^{c \times h \times w}$
- Augmented views: x_i, x'_i
- Embeddings: $K, K' \in R^{M \times N}$
  - $where\ K_{\cdot,i} = p_\theta\big(f_\theta(x_i)\big), K'_{\cdot,i} = p_\theta\big(f_\theta(x'_i)\big)$



Notations & Definitions

Definitions

Definition 3.1. Given a matrix $A \in \mathbb{R}^{n \times n}$. We define its *extracted diagonal* $\text{diag}(A) \in \mathbb{R}^{n \times n}$ as:

$$\text{diag}(A)_{i,j} = \begin{cases} A_{i,i}, & \text{if } i = j \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Definition 3.2. A method is said to be *sample-contrastive* if it minimizes the contrastive criterion $L_c = \|\mathcal{K}^T\mathcal{K} - \text{diag}(\mathcal{K}^T\mathcal{K})\|_F^2$. Similarly, a method is said to be *dimension-contrastive* if it minimizes the non-contrastive criterion $L_{nc} = \|\mathcal{K}\mathcal{K}^T - \text{diag}(\mathcal{K}\mathcal{K}^T)\|_F^2$.

Invariance criterion. While L_c and L_{nc} focus on regularizing the embedding space, they are not optimized alone. They are usually combined with an invariance criterion that aims at producing the same embedding for two views of the same image. As such, a complete self-supervised loss would look like $L_{SSL} = L_{inv} + L_{reg}$ with L_{reg} being either L_c or L_{nc} for our prototypical sample-contrastive and dimension-contrastive methods.

Why SimCLR is a Sample-Contrastive Method?

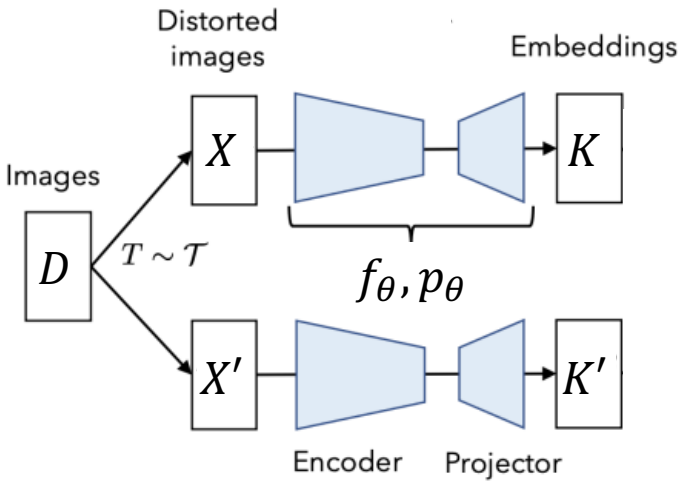
SimCLR-sq/abs: We now take a look at SimCLR-abs/sq's criteria. We consider that $\mathcal{K}$ is l2 normalized column-wise, i.e. embeddings are normalized. Let $f : \mathbb{R} \to \mathbb{R}^+$ be either defined as $f(x) = x^2$ for SimCLR-sq or as $f(x) = |x|$ for SimCLR-abs. We have

$$\mathcal{L}_{\text{SimCLR}} = \sum_{i=1}^{N} -\log\left(\frac{e^{f\left(\mathcal{K}_{\cdot,i}^T \mathcal{K}'_{\cdot,i}\right)/\tau}}{e^{f\left(\mathcal{K}_{\cdot,i}^T \mathcal{K}'_{\cdot,i}\right)/\tau} + \sum_{j\neq i} e^{f\left(\mathcal{K}_{\cdot,i}^T \mathcal{K}_{\cdot,j}\right)/\tau}}\right) \tag{30}$$

$$= \sum_{i=1}^{N} -\frac{f\left(\mathcal{K}_{\cdot,i}^T \mathcal{K}'_{\cdot,i}\right)}{\tau} + \log\left(e^{f\left(\mathcal{K}_{\cdot,i}^T \mathcal{K}'_{\cdot,i}\right)/\tau} + \sum_{j\neq i} e^{f\left(\mathcal{K}_{\cdot,i}^T \mathcal{K}_{\cdot,j}\right)/\tau}\right). \tag{31}$$

Nonetheless one can clearly see that to minimize this criterion, all the similarities between the negative pairs, i.e. $\forall i, \forall j \neq i,\ f\left(\mathcal{K}_{\cdot,i}^T \mathcal{K}_{\cdot,j}\right)$, need to be minimized. As this will result in a diagonal Gram matrix, we can say that minimizing this criterion will also minimize our sample-contrastive one. We can thus conclude that SimCLR-sq and SimCLR-abs are sample-contrastive methods.

- Empirically, performance of $L_{SimCLR-sq}$ is identical to $L_{SimCLR}(f(x) = x)$.
- Minimizing $L_{SimCLR-sq}$ is identical to minimizing $L_{inv} + L_{sc}$.
- Thus, SimCLR can be interpreted as a Sample-Contrastive method.



Why VICReg is a Dimension-Contrastive Method?

VICReg: VICReg's criterion is defined as

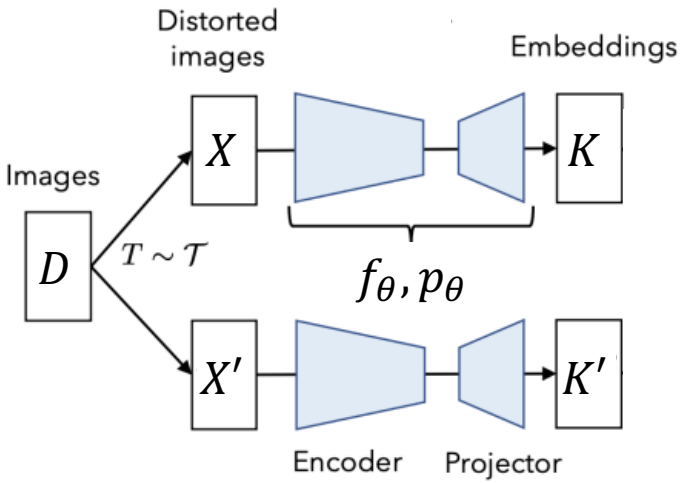
$$\mathcal{L}_{VICReg} = \lambda \sum_{i=1}^{N} \|\mathcal{K}_{\cdot,i} - \mathcal{K}'_{\cdot,i}\|_2^2 + \mu\left(v(\mathcal{K}) + v(\mathcal{K}')\right) + \nu\left(c(\mathcal{K}) + c(\mathcal{K}')\right). \tag{20}$$

With c a criterion that penalizes the off-diagonal terms of the covariance matrix as

$$c(\mathcal{K}) = \sum_{i \neq j} \mathrm{Cov}(\mathcal{K})_{i,j}^2 = \|\mathcal{K}\mathcal{K}^T - \mathrm{diag}(\mathcal{K}\mathcal{K}^T)\|_F^2 = L_{nc}, \tag{21}$$

and v a criterion that aims at normalizing dimensions, i.e. rows of $\mathcal{K}$.

- Minimizing L_{VICReg} is identical to minimizing $L_{inv} + L_{nc}$.
- Thus, VICReg can be interpreted as a Dimension-Contrastive method.



Summary

- SimCLR is a sample-contrastive method with L_2 normalized embeddings.
- VICReg is a dimension-contrastive method with L_2 normalized dimensions (minimizing $v(K)$).

Equivalence of Sample-contrast and Dimension-contrast

Relationship between sample-contrast and dimension-contrast

Theorem 3.3. The sample-contrastive and dimension-contrastive criteria L_c and L_{nc} are equivalent up to row and column normalization of the embedding matrix $\mathcal{K}$. Consider a batch size of N and an embedding dimension of M. We have:

$$L_{nc} + \sum_{j=1}^{M} \|\mathcal{K}_{j,\cdot}\|_2^4 = L_c + \sum_{i=1}^{N} \|\mathcal{K}_{\cdot,i}\|_2^4. \tag{3}$$

Let $K = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

- $L_{nc} = \|KK^T - \mathrm{diag}(KK^T)\|_F^2$

$$= \left\| \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} - \mathrm{diag}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} \right) \right\|_F^2$$

$$= \left\| \begin{bmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{bmatrix} - \begin{bmatrix} a^2+b^2 & 0 \\ 0 & c^2+d^2 \end{bmatrix} \right\|_F^2$$

$$= \left\| \begin{bmatrix} 0 & ac+bd \\ ac+bd & 0 \end{bmatrix} \right\|_F^2$$

$$= (ac+bd)^2 + (ac+bd)^2$$

$$= 2(ac+bd)^2.$$

- $L_c = 2(ab+cd)^2.$
- $L_{nc} - L_c = 2\{(ac+bd)^2 - (ab+cd)^2\}$

$$= 2(a^2c^2 + b^2d^2 - a^2b^2 - c^2d^2).$$

- $\sum_{j=1}^{N} \|K_{\cdot,i}\|_2^4 = (a^2+c^2)^2 + (b^2+d^2)^2.$
- $\sum_{j=1}^{M} \|K_{i,\cdot}\|_2^4 = (a^2+b^2)^2 + (c^2+d^2)^2.$
- $\sum_{j=1}^{N} \|K_{\cdot,i}\|_2^4 - \sum_{i=1}^{M} \|K_{i,\cdot}\|_2^4 = 2(a^2c^2 + b^2d^2 - a^2b^2 - c^2d^2).$

$$\therefore L_{nc} - L_c = \sum_{i=1}^{N} \|K_{\cdot,i}\|_2^4 - \sum_{j=1}^{M} \|K_{i,\cdot}\|_2^4.$$

- A proof is on the appendix.

Equivalence of Sample-contrast and Dimension-contrast

Relationship between sample-contrast and dimension-contrast

Theorem 3.3. The sample-contrastive and dimension-contrastive criteria L_c and L_{nc} are equivalent up to row and column normalization of the embedding matrix $\mathcal{K}$. Consider a batch size of N and an embedding dimension of M. We have:

$$L_{nc} + \sum_{j=1}^{M} \|\mathcal{K}_{j,\cdot}\|_2^4 = L_c + \sum_{i=1}^{N} \|\mathcal{K}_{\cdot,i}\|_2^4. \tag{3}$$

terchangeably. However, if both rows and columns of $\mathcal{K}$ were L2 normalized, we would have $L_{nc} = L_c + N - M$. In this case, both criteria would be equivalent from an optimization point of view, and we could conclude that sample-contrastive and dimension-contrastive methods are all minimizing the same criterion.

- $\|K_{j,\cdot}\|_2^4$: a sum of the norm of the features.
- $\|K_{\cdot,i}\|_2^4$: a sum of the norm of the embeddings.
- Therefore, if both rows and columns of K are normalized,

  sample-contrastive methods and dimension-contrastive methods are minimizing the same criterion.

Equivalence of Sample-contrast and Dimension-contrast

Relationship between sample-contrast and dimension-contrast

Lemma 3.4. If embeddings are normalized such that $\forall i,\ \|\mathcal{K}_{\cdot,i}\|_2 = a$ we have

$$\frac{N^2}{M}a^4 \leq \sum_{j=1}^{M} \|\mathcal{K}_{j,\cdot}\|_2^4 \leq N^2 a^4. \tag{4}$$

Conversely, if dimensions are normalized such that $\forall j,\ \|\mathcal{K}_{j,\cdot}\|_2 = a$ we have

$$\frac{M^2}{N}a^4 \leq \sum_{i=1}^{N} \|\mathcal{K}_{\cdot,i}\|_2^4 \leq M^2 a^4. \tag{5}$$

- In SimCLR, embeddings are L_2 normalized.
- In VICReg, dimensions are L_2 normalized.

Corollary 3.4.1. If embeddings are L2-normalized we have

$$L_{nc} - N + \frac{N^2}{M} \leq L_c \leq L_{nc} - N + N^2. \tag{6}$$

Similarly, if dimensions are L2-normalized we have

$$L_c - M + \frac{M^2}{N} \leq L_{nc} \leq L_c - M + M^2. \tag{7}$$

Summary

- If we minimize L_c with L_2-normalized embeddings, L_{nc} cannot be arbitrarily large.
- If we minimize L_{nc} with L_2-normalized dimensions, L_c cannot be arbitrarily large.

- If we minimize L_{SimCLR}, L_{VICReg} cannot be arbitrarily large.
- If we minimize L_{VICReg}, L_{SimCLR} cannot be arbitrarily large.

Interpolating between VICReg and SimCLR

$$\text{VICReg} \xrightarrow{\text{LogSumExp}} \text{VICReg-exp} \xrightarrow{\text{Contrastive}} \text{VICReg-ctr} \xrightarrow{\text{Neg. pair sampling}} \text{SimCLR}$$

$$\mathcal{L}_{VICReg-exp} = \lambda \sum_{i=1}^{N} \|\mathcal{K}_{\cdot,i} - \mathcal{K}'_{\cdot,i}\|_2^2 + \mu \left(v(\mathcal{K}) + v(\mathcal{K}')\right) + \nu \left(c_{exp}(\mathcal{K}) + c_{exp}(\mathcal{K}')\right). \quad (12)$$

$$\mathcal{L}_{VICReg-ctr} = \lambda \sum_{i=1}^{N} \|\mathcal{K}_{\cdot,i} - \mathcal{K}'_{\cdot,i}\|_2^2 + \mu \left(v(\mathcal{K}^T) + v(\mathcal{K}'^T)\right) + \nu \left(c_{exp}(\mathcal{K}^T) + c_{exp}(\mathcal{K}'^T)\right). \quad (13)$$

$$c_{exp}(\mathcal{K}) = \frac{1}{d} \sum_i \log \left(\sum_{j \neq i} e^{C(\mathcal{K})_{i,j}/\tau} \right)$$

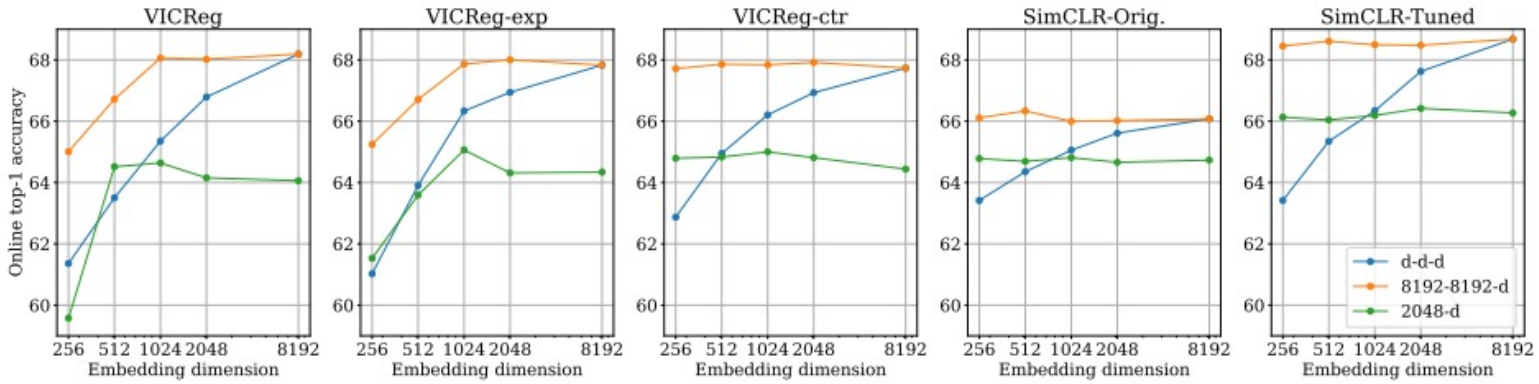


Figure 1: VICReg, VICReg-exp, and VICReg-ctr perform similarly in 100 epochs training, validating empirically our theoretical result. While the original implementation of SimCLR performs significantly worse – which is unexpected per our theory – we are able to improve its performance to VICReg's level. This further validates our findings. While different projector architectures impact performance, behaviors are similar across methods. Confer supplementary section K for numerical

Connections to other methods

	L_{inv}	L_c	L_{nc}	Predictor + stop.grad	Momentum encoder	Clustering (+ m-crop)	IN-1K linear
-	✓						collapse
SimCLR [1]	✓	✓					69.3
MoCo [2]	✓	✓			✓		71.1
VICReg [3]	✓		✓				73.2
SimSiam [4]	✓			✓			71.3
BYOL [5]	✓			✓	✓		74.3
SWaV [6]	✓	✓			✓	✓	75.3
DINO [7]	✓	✓			✓	✓	75.3

- At least one regularization objective is required (L_c, L_{nc} or predictor + stop.grad).
- According to [8],[9], when predictor is one linear layer, predictor + stop.grad is implicitly performing dimension contrast.
- Using momentum encoder is generally helpful.
- Clustering + multi-crop can be also helpful.

Conclusion

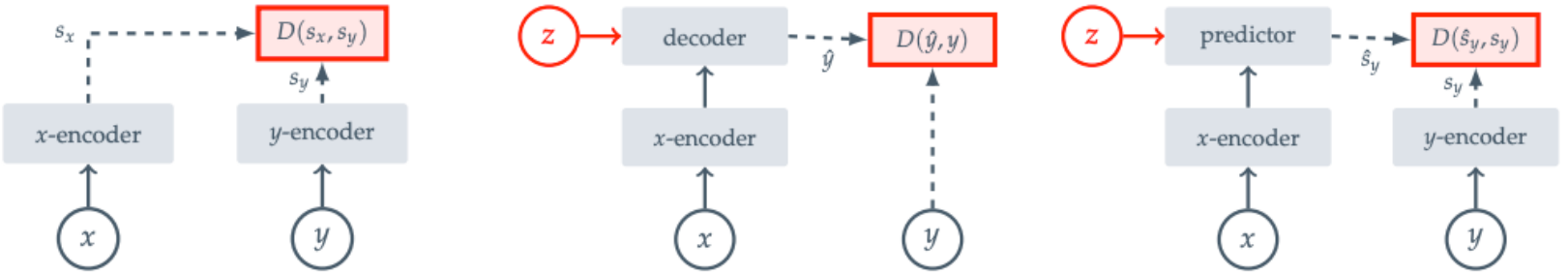
Summary

- **Sample-contrastive methods and dimension-contrastive methods are two sides of the same coin.**
- Using predictor + stop.grad can be interpreted as using a dimension-contrastive regularization with an architectural design.

Practical advice (personal experience)

- L_c: stable, perform well with a large batch size.
- L_{nc}: stable, perform well with a large dimension (can be a good choice in a large model).
- Predictor + stop.grad: unstable, but generally better upper bound.
- Momentum Encoder: stable, helpful when you train with long epochs.
- Clustering + Multi-Crop: relatively hard to implement (various tricks), but better upper bound.

Future Work



(a) **Joint-Embedding Architecture** (b) **Generative Architecture** (c) **Joint-Embedding Predictive Architecture**

Method	Arch.	Epochs	Top-1
Methods without view data augmentations			
data2vec [7]	ViT-L/16	1600	77.3
MAE [35]	ViT-B/16	1600	68.0
	ViT-L/16	1600	76.0
	ViT-H/14	1600	77.2
CAE [21]	ViT-B/16	1600	70.4
	ViT-L/16	1600	78.1
I-JEPA	ViT-B/16	600	72.9
	ViT-L/16	600	77.5
	ViT-H/14	300	79.3
	ViT-H/16_{448}	300	**81.1**
Methods using extra view data augmentations			
SimCLR v2 [20]	RN152 (2×)	800	79.1
DINO [17]	ViT-B/8	300	80.1
iBOT [75]	ViT-L/16	250	**81.0**

Table 1. **ImageNet**. Linear-evaluation on ImageNet-1k (the ViT-H/16_{448} is pretrained at at a resolution of 448 × 448). I-JEPA improves linear probing performance compared to other methods that do not rely on hand-crafted view data-augmentations during pre-training. Moreover, I-JEPA demonstrates good scalability — the larger I-JEPA model matches the performance of view-invariance approaches without requiring view data-augmentations.

Integrating Generative Modeling & Invariance Learning

- Invariance Learning (Joint-Embedding architecture)
  - (Cons) Augmentation is hand-crafted.
- Generative Modeling (Masked Image Modeling)
  - (Cons) Requires to reconstruct pixels.
- Invariance + Generative Modeling